

Experiment 33

Aim

To perform morphological gradient operation on an image using OpenCV.

Procedure

1. Import the required libraries, OpenCV and NumPy.
2. Read the input image.
3. Convert the image into grayscale.
4. Create a morphological kernel.
5. Apply the Morphological Gradient operation using cv2.morphologyEx().
6. Display the original and gradient images.
7. Save the resulting image..

Code:

```
import cv2

import numpy as np

img = cv2.imread("image.jpg")

if img is None:

    print("Error: Image not found!")

    exit()

gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)

kernel = np.ones((5, 5), np.uint8)

gradient = cv2.morphologyEx(gray, cv2.MORPH_GRADIENT, kernel)

cv2.imshow("Original Image", img)

cv2.imshow("Morphological Gradient", gradient)
```

```
cv2.imwrite("morphological_gradient.jpg", gradient)
```

```
cv2.waitKey(0)
```

```
cv2.destroyAllWindows()
```

Sample Input:



Output:



Result

The morphological gradient operation was successfully performed using OpenCV. The operation highlighted the boundaries and edges of objects in the image.